Science Team Weekly

Science literacy in action: students read closely, represent evidence, and explain their thinking.

THIS WEEK'S DEPARTMENT LENS

Literacy Is How Students Make Their Science Thinking Visible

Across Biology and Pre-AP Biology, students used structured reading, data tables, physical models, discussion, and written explanation to turn information into evidence-based scientific ideas. These are not add-ons to science instruction—they are the practices that help students think like scientists.

01 READ

Locate the central scientific idea, supporting evidence, and unfamiliar academic language before responding.

02 REPRESENT

Use data tables, diagrams, and models to show relationships that are difficult to see in words alone.

03 REASON & REVISE

Explain a claim using evidence, receive feedback, and improve the explanation with greater precision.

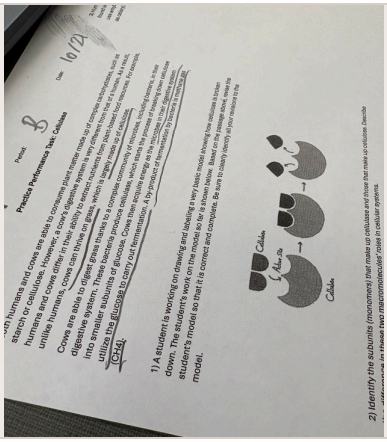
CLOSE READING MEETS MODEL-BASED THINKING

From Cellulose to a Stronger Explanation

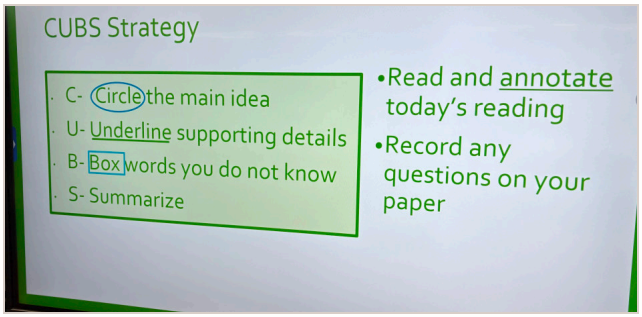
In **Mr. Robert Kolbeck's Pre-AP Biology class**, students used the CUBS routine to read and annotate a text about why cows can digest cellulose while humans cannot. The routine asked students to **Circle** the main idea, **Underline** supporting details, **Box** unfamiliar words, and **Summarize** before they moved into model-based questions.

Students then connected text evidence to a model of enzyme action: cellulase helps break down cellulose, while human amylase acts on starch—not cellulose. By pairing annotation with model revision and discussion, the task positioned scientific vocabulary as a tool for reasoning, not simply a list to memorize.

> *"Read closely, identify evidence, and use the model to explain why."*



Scientific reading in action: *a student task links annotation, vocabulary, model revision, and explanation.*



CUBS: *a repeatable routine for finding meaning in a science text.*

FROM TEXT TO EVIDENCE-BASED EXPLANATION

Find it: Which sentence or model feature gives usable evidence? **Name it:** What science word makes the idea more precise? **Explain it:** How does that evidence support the claim about enzyme specificity?

VOCABULARY THAT CARRIES THE EXPLANATION

Cellulose: a plant carbohydrate. **Cellulase:** an enzyme that breaks cellulose into smaller sugars. **Specificity:** the match between an enzyme and its substrate. Students can use these terms to move from a general observation to a precise biological explanation.

SHARED LITERACY GOAL

Students should be able to read a scientific source, identify usable evidence, and explain how that evidence supports a scientific claim. The same habit transfers from enzymes and osmosis to ecosystems, chemistry, and beyond.

STUDENT SELF-CHECK

Before I submit: Can I identify the main idea, point to evidence, explain the model, and use the vocabulary precisely?

TEACHER LOOK-FOR

Listen for students using the text, data, or model to explain—not simply repeat—a scientific idea.

From Measurements to Meaning

Pre-AP Biology students use data as evidence in an osmosis investigation.

PRE-AP BIOLOGY · MR. KEVIN FLYNN

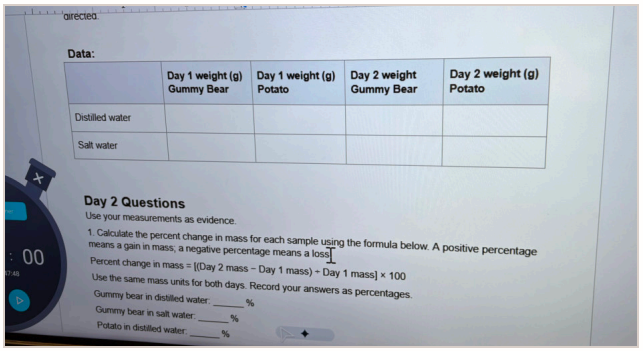
Osmosis Requires More Than a Prediction

In an osmosis investigation, students gathered measurements for gummy bears and potato samples placed in distilled water and salt water. They recorded Day 1 and Day 2 masses, calculated percent change in mass, and used the pattern to make sense of water movement across a membrane.

The strength of this kind of task is the **evidence trail**: students must keep units consistent, compare conditions, calculate a quantitative change, and use the numbers to support—not replace—their explanation. The data table becomes a draft of the scientific argument.

A DATA-LITERACY ROUTINE

- **Record carefully:** capture the measurement and the condition.
- **Calculate:** use percent change to compare samples fairly.
- **Analyze:** notice patterns across distilled and salt water.
- **Explain:** connect the pattern to the direction of water movement.


directed.

Data:

	Day 1 weight (g) Gummy Bear	Day 1 weight (g) Potato	Day 2 weight Gummy Bear	Day 2 weight (g) Potato
Distilled water				
Salt water				

Day 2 Questions

Use your measurements as evidence.

1. Calculate the percent change in mass for each sample using the formula below. A positive percentage means a gain in mass; a negative percentage means a loss.

Percent change in mass = [(Day 2 mass − Day 1 mass) ÷ Day 1 mass] × 100

Use the same mass units for both days. Record your answers as percentages.

Gummy bear in distilled water: _____ %

Gummy bear in salt water: _____ %

Potato in distilled water: _____ %

Evidence in progress: *students record Day 1 and Day 2 mass and calculate percent change before explaining the result.*

LANGUAGE THAT MAKES REASONING VISIBLE

"The mass **increased/decreased** in ___ solution. This pattern suggests water moved ___. The data support this because ___."

Literacy move: *Ask students to point to a value, a comparison, and a calculation before asking them to explain a scientific process.*

WHY IT MATTERS

Data Analysis Is Reading, Writing, and Reasoning

Students are often asked to "use evidence," but a data set can feel disconnected from the explanation that follows. Naming the sequence—measurement, calculation, comparison, claim—helps students see how a number earns its role as evidence.

In the osmosis investigation, the percent-change calculation gives students a common language for comparing samples. From there, they can construct a more precise explanation about concentration gradients and water movement.

QUESTIONS THAT STRENGTHEN THE EXPLANATION

What changed? Which sample gained or lost mass?

What comparison matters? How do the two solutions differ?

What does that pattern mean? What does it suggest about the direction of osmosis?

How do you know? Which values and calculations support the claim?

EVIDENCE-BASED SCIENCE WRITING

When students interpret numbers in context, they move from reporting a result to explaining a biological mechanism.

DATA CHECK

Can the reader find the value? Name the condition, include a number or calculated change, and compare it to another result.

REASONING CHECK

Can the reader follow the mechanism? Link the data pattern to water movement and the relevant concentration difference.

FROM CALCULATION TO CLAIM

1. STATE THE PATTERN

"The sample in ___ changed by ___%."

2. COMPARE CONDITIONS

"This is greater/less than the sample in ___."

3. EXPLAIN THE MECHANISM

"This supports the idea that water moved ___ because ___."

Reading an Ecosystem, Building a Model

Biology students connect an annotated article to a food-web explanation.

BIOLOGY · MR. KEVIN FLYNN & MS. CYKKA LOVETT

Food-Web Puzzles Make Relationships Visible

Students built a Cape Cod marine food web by arranging organism cards and drawing arrows to represent feeding relationships. Phytoplankton, zooplankton, sand lance, striped bass, grey seals, seagulls, and great white sharks became a model students could test, revise, and explain.

With **Mr. Kevin Flynn and co-teacher Ms. Cykka Lovett** supporting the work, students used the model to predict how population changes could cascade through the ecosystem if a keystone predator were removed. The activity made key academic terms—*predator, prey, population, keystone species, increase, decrease,* and *trophic cascade*—purposeful tools for explaining an ecological pattern.

MODEL-TO-EXPLANATION PROMPT

Claim: If the great white shark were removed, ___ would increase/decrease.

Evidence: In our food-web model, the arrows show that ___.

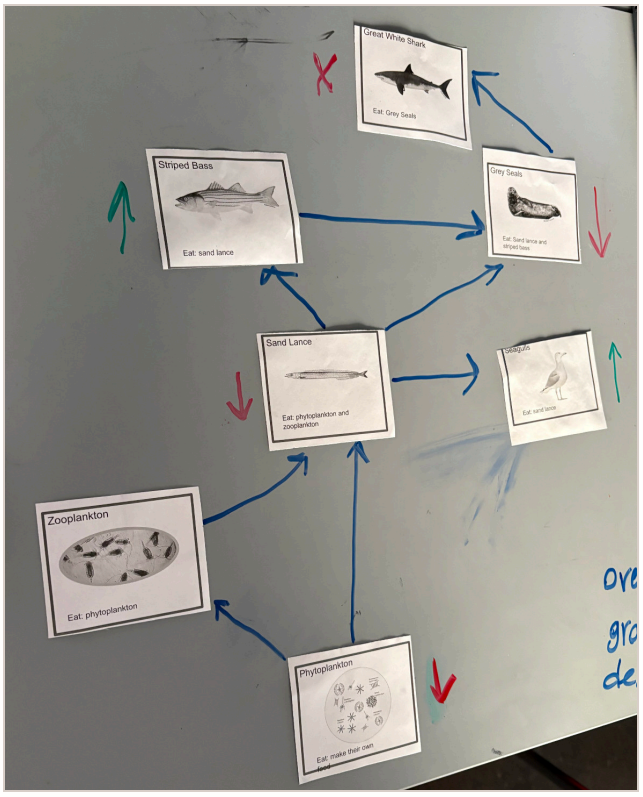
Reasoning: Because ___ eats ___, the population change creates a trophic cascade by ___.

READ THE ARROWS

Ask students to trace one pathway aloud: “Energy moves from ___ to ___. If this population changes, ___ may change because ___.” This turns a visual model into a rehearsed scientific explanation.

PREDICT, TEST, REVISE

Predict: What changes first? **Test:** Trace the arrows that support or challenge the idea.
Revise: Adjust the claim when the model reveals a more complete relationship.



Modeling a trophic cascade: *students use organism cards and arrows to make the system’s relationships visible.*

CLOSE READING IN BIOLOGY · MR. NATHANIEL WALKER

Annotation Prepares Students to Think with a Text

In **Mr. Nathaniel Walker’s Biology class**, students read and annotated an article about the great white shark. Annotation slows down the reading process in a productive way: students identify a main idea, locate details, notice unfamiliar words, and record questions before being asked to explain the science.

That process is especially valuable in ecology, where students need to trace relationships across a text and then transfer evidence into a diagram, a prediction, or a written explanation.

ANNOTATION MOVES FOR A SHARK ARTICLE

- Circle the central claim about the shark’s role in the ecosystem.
- Underline details that explain predator–prey relationships.
- Box terms such as *apex predator* and *keystone species*.
- Write one question about a possible population change.
- Summarize the evidence before building or revising a model.

LITERACY ACROSS REPRESENTATIONS

A strong science explanation asks students to carry an idea from **text → model → discussion → writing**, refining the same claim as the evidence becomes clearer.

Marine Debris Forensics Comes to NBHS

A local marine-science visit invites students to ask how evidence can tell a story about the ocean.

GUEST SCIENTIST & EDUCATOR SPOTLIGHT

Thank You, Ms. Michelle Carvalho

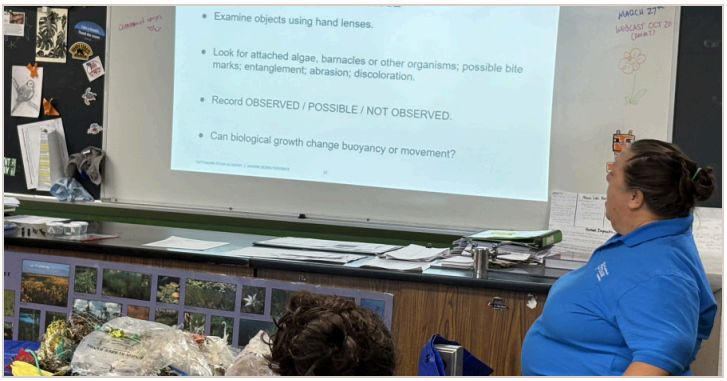
Ms. Michelle Carvalho, Program Director at Cuttyhunk STEAM Academy, visited AP Environmental Science and Marine Science classes to share a presentation on marine debris forensics. The experience connected classroom learning to real questions about evidence, ecosystems, materials, and the human choices that shape coastal environments.

Marine debris forensics uses observation and analysis to investigate where debris may have come from, how it moved through the environment, and what its presence can reveal about human activity and coastal systems. It invites students to treat a material artifact as evidence rather than simply as "trash."

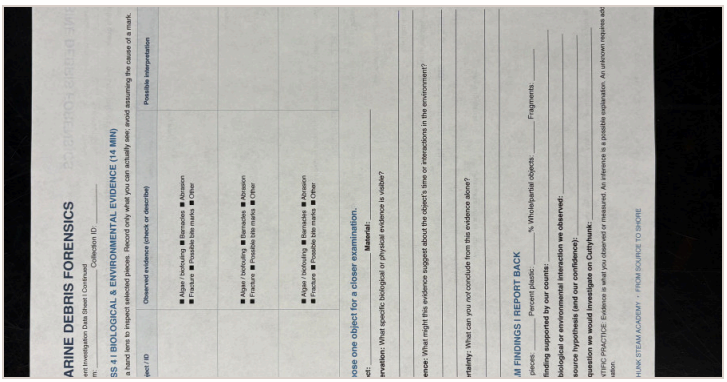
For AP Environmental Science and Marine Science students, this kind of inquiry brings together scientific literacy skills: **observe carefully, describe precisely, ask questions, compare evidence, and communicate a claim with appropriate limits.**

QUESTIONS A MARINE DEBRIS INVESTIGATOR CAN ASK

- What material is this, and what properties help identify it?
- What clues suggest where it may have originated or how long it has been in the marine environment?
- How might organisms, currents, weather, or human activity affect its movement?
- What additional evidence would strengthen or challenge an explanation?



Connecting science to place: *students explore marine debris through an evidence-centered local lens.*



Questions drive inquiry: *a community-science presentation expands classroom discussion.*

Science literacy beyond the textbook: *"What do we notice? What might it mean? What evidence would we need next?"*

A BRIDGE BETWEEN COURSES

OBSERVE
Use descriptive language to record visible features, patterns, and possible sources.

INTERPRET
Connect observations to concepts such as materials, transport, food webs, and human impacts.

COMMUNICATE
Share a claim with evidence while naming what remains uncertain or needs further investigation.

THANK YOU
We appreciate **Ms. Carvalho and Cuttyhunk STEAM Academy** for bringing a relevant, place-based scientific question to our students and strengthening the connection between literacy, inquiry, and the marine environment.

The Dynamic Genome

How epigenetics, environmental signals, and biological clocks help regulate gene expression.

SCIENCE CORNER · EPIGENETICS

Same DNA Sequence. Different Instructions in Use.

Nearly every cell in a person carries the same DNA sequence, yet a neuron, muscle cell, and skin cell do very different jobs. One reason is **gene regulation**: cells control which genes are accessible, when they are used, and how much protein is made. Epigenetics describes heritable changes in gene expression that do not change the DNA sequence itself. DNA methylation, chemical changes to histones, and the packaging of DNA into chromatin are part of that regulation [1, 2].

It is more accurate to say that our genome is **dynamically regulated** than that it is constantly rewritten. Cells respond to signals from inside and outside the body, but the DNA letters generally remain the same. Environmental exposures, behavior, development, aging, metabolism, and cell-to-cell signals can influence gene-expression patterns. These relationships are complex and context dependent—not every exposure produces the same effect in every person or cell [1, 2].

CORE DISTINCTION

Genetic change alters the DNA sequence. **Epigenetic change** influences how the cell reads and uses that sequence. Both can matter in health and disease.



Conceptual view: *molecular signals, chromatin regulation, and biological timing help shape which genes are active in a particular cell.*

THINK LIKE A SCIENTIST

If two cells share the same DNA, what evidence would help explain why they produce different proteins?

INTERNAL TIMING: THE BIOLOGICAL CLOCK

Gene Activity Also Has a Rhythm

The body's circadian system helps coordinate daily patterns in gene expression, metabolism, hormones, and cell behavior. Clock proteins and chromatin regulators work together, so some gene-expression programs rise and fall across the day. Research describes rhythmic changes in histone modifications and clock-controlled gene activity rather than a simple, universal "on/off switch" [3].

That makes timing a valuable scientific lens: a cell must integrate its identity, current signals, energy state, and biological time when regulating a gene. The result is a responsive system, not a fixed script.

ENVIRONMENTAL AND INTERNAL SIGNALS

External: nutrition, toxicants, stressors, infections, and other exposures can be associated with changes in epigenetic regulation.

Internal: developmental stage, hormones, metabolism, cell communication, and circadian timing also help coordinate gene activity.

Association does not mean a single factor alone determines disease. Biology reflects interacting genetic, epigenetic, environmental, and social influences.

CANCER CONNECTION

In cancer, abnormal DNA methylation patterns and chromatin changes can disrupt normal control of gene expression. These changes may silence genes that help restrain growth or support programs that promote tumor behavior. Epigenetic changes are one part of a multifactorial disease process and are being studied for diagnosis and treatment alongside genetic alterations [1, 4, 5].

SCIENCE CORNER VOCABULARY

Gene expression: using genetic instructions to make a functional product. **Chromatin:** DNA packaged with proteins. **DNA methylation:** a chemical mark that can influence gene activity. **Circadian rhythm:** an approximately daily pattern in biological processes.

SELECTED REFERENCES

[1] CDC. *Epigenetics, Health, and Disease* (2025). https://www.cdc.gov/genomics-and-health/epigenetics/index.html

[2] Jaenisch, R. & Bird, A. *Epigenetic regulation of gene expression: how the genome integrates intrinsic and environmental signals.* Nature Genetics 33, 245–254 (2003). https://doi.org/10.1038/ng1089

[3] Masri, S. & Sassone-Corsi, P. *Circadian clocks, epigenetics, and cancer.* Current Opinion in Oncology 27, 50–56 (2015). https://pmc.ncbi.nlm.nih.gov/articles/PMC4732884/

[4] Flavahan, W. A., Gaskell, E. & Bernstein, B. E. *Epigenetic plasticity and the hallmarks of cancer.* Science 357 (2017). https://doi.org/10.1126/science.aal2380 [5] NCI. *Epigenetics SPOREs* (2025). https://dctd.cancer.gov/research/spores/focus-area/epigenetics/introtoepigenetics

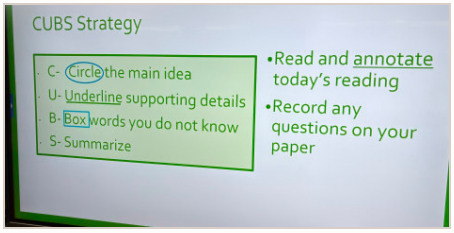
Language Supports That Protect Scientific Rigor

Multiple ways for students to access a complex idea, rehearse language, and explain their thinking.

SHARED RESOURCE SPOTLIGHT

Making Science Understandable—and Explainable

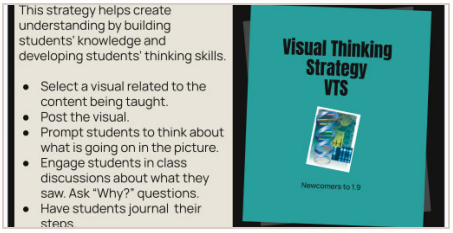
Science literacy grows when students can enter a challenging idea through visuals, discussion, annotation, and structured writing. The routines below support multilingual learners and benefit every student by making disciplinary language, evidence, and reasoning more visible.



CUBS strategy: *circle, underline, box, summarize.*

1. CUBS ANNOTATION

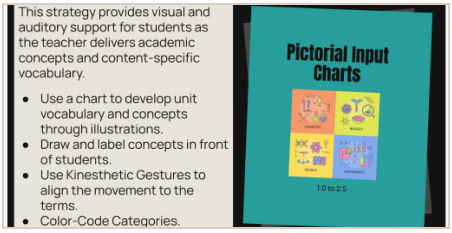
Ask students to **Circle** the main idea, **Underline** supporting details, **Box** words they do not know, and **Summarize** the reading. Invite them to record questions before they write.



Visual Thinking Strategy: *observe, discuss, and journal.*

2. VISUAL THINKING

Post a course-aligned visual and ask: **"What is happening?"** and **"What makes you say that?"** Give students time to talk, then journal the steps in their reasoning.



Pictorial input charts: *vocabulary connected to meaning.*

3. PICTORIAL INPUT CHARTS

Draw and label unit concepts as you teach. Use illustrations, gestures, and color-coded categories to make vocabulary concrete and keep the language connected to the phenomenon.

A PRACTICAL PLANNING LENS

Keep the Task Rigorous. Make the Path Visible.

Language support does not mean reducing the scientific thinking students are asked to do. Instead, it means planning a way into the same grade-level task. A visual, word bank, annotated source, structured partner conversation, sentence frame, or modeled response can help students access the evidence they need to reason.

BEFORE LEARNING
Preview key vocabulary with an image, gesture, or pictorial chart.

DURING LEARNING
Use CUBS or a similar annotation routine to make the text and evidence manageable.

AFTER LEARNING
Give students a structured way to explain, revise, and extend their scientific thinking.

TRY ONE LITERACY ROUTINE NEXT WEEK

For a reading or stimulus: Use CUBS before discussion or writing.

For a diagram, data display, or phenomenon: Begin with "What do you notice?" and "What evidence supports that?"

For vocabulary: Add a labeled image, gesture, or color-coded category before asking students to use the term in explanation.

For a written response: Offer a short evidence-and-reasoning sentence frame, then make time for students to revise one part of the response.

DEPARTMENT GRATITUDE

Special shoutout to our **custodial team** for the hard work of delivering instructional materials to Science classrooms.

Thank you to **Ms. Hope Bynoe** and **Ms. Brittany Jenney** for providing these ELD instructional strategies for our department to study and use.

We also appreciate every colleague who shares curriculum resources across **Biology, Chemistry, and Environmental Science** so all team members are supported.

Thank you to everyone who continues to offer feedback and support during PLCs and PDs. Your partnership helps me grow as a leader and strengthens our shared learning experience.

DEPARTMENT COMMITMENT

We will continue to build classrooms where every student can **read science, talk science, model science, and write science**—with the support needed to participate fully in rigorous learning.

A COLLABORATIVE PLANNING CHECK

PLAN THE STUDENT EXPERIENCE

Before the lesson, name what students will **see, say, read, write, and revise**. Then decide which visual, language, or discussion support helps them access the same rigorous science task.

LOOK FOR EVIDENCE OF ACCESS

During and after the lesson, collect one signal: an annotation, partner explanation, model label, data claim, or revision that shows how students used the support to strengthen their thinking.

Analyzing Evidence in Chemistry

High-quality materials, analytical thinking, and meaningful science learning.

SCIENCE TEAM SHOUTOUT

Thank You, Ms. Margaret Henning

Special recognition to **Ms. Margaret Henning** for the great care and effort she brings to implementing **OpenSciEd Chemistry and Pre-AP Chemistry**. Her dedication to high-quality instructional materials, collaboration with colleagues, and support for student analytical thinking is clearly strengthening science learning.

FROM DATA DISPLAY TO SCIENTIFIC QUESTION

Students Practice Reading Information Like Scientists

In Chemistry learning experiences, students encounter visual information such as charts, tables, and data displays. Rather than simply noticing a result, they are asked to interpret what the information shows, identify patterns, and generate questions that can guide further investigation.

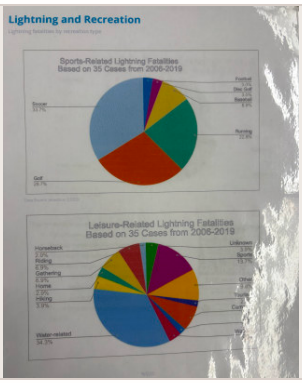
The lightning-and-recreation data display is an example of a meaningful analytical prompt. Students can compare categories, notice variation, consider what information is missing, and decide what evidence would be needed to support a conclusion. This work builds the habits students need to analyze data with precision.

ANALYTICAL THINKING MOVES

Notice: What pattern, difference, or outlier stands out?

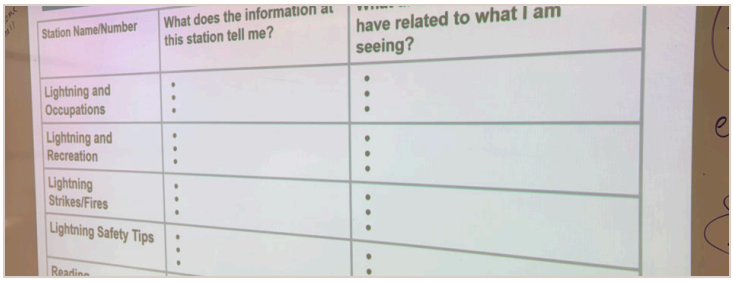
Interpret: What does the information suggest—and what does it not prove?

Question: What additional data or context would strengthen the explanation?



Data as a text: *students use charts to identify patterns, make comparisons, and form questions.*

Literacy in Chemistry: *Students learn to move from "I see a chart" to "The data suggest ___ because ___; I still need to know ___."*



Organizing evidence: *a notebook structure helps students record what a station tells them and the questions that remain.*

HIGH-QUALITY, RELEVANT MATERIALS

Rigor and Relevance Work Together

Ms. Henning's work helps students practice analyzing evidence while engaging with learning materials that are relevant to their lives and communities. Culturally relevant examples invite students to see science as a way to understand the world around them, while rigorous tasks ask them to reason carefully with evidence.

Her willingness to work alongside colleagues and share resources strengthens both OpenSciEd Chemistry and Pre-AP Chemistry. This collaboration helps ensure that students have access to coherent, standards-aligned learning experiences and the support needed to succeed.

DEPARTMENT APPRECIATION

Thank you, **Ms. Margaret Henning**, for your commitment to high-quality Chemistry instruction, analytical literacy, culturally relevant materials, and generous collaboration in support of colleagues and students.

WHAT STUDENTS PRACTICE

ANALYZE

Compare categories, quantify differences, and identify a pattern or outlier in a visual data source.

QUESTION

Distinguish what the information suggests from what it cannot yet answer, then name useful next evidence.

COMMUNICATE

Use precise language to support a conclusion and connect the scientific question to a relevant context.

STUDENT DISCOURSE STARTERS

"I notice ___." "The evidence that supports this is ___." "A question I still have is ___." "To investigate further, I would need ___."